Discrete-Time Markov Chains (DTMCs): Part 1

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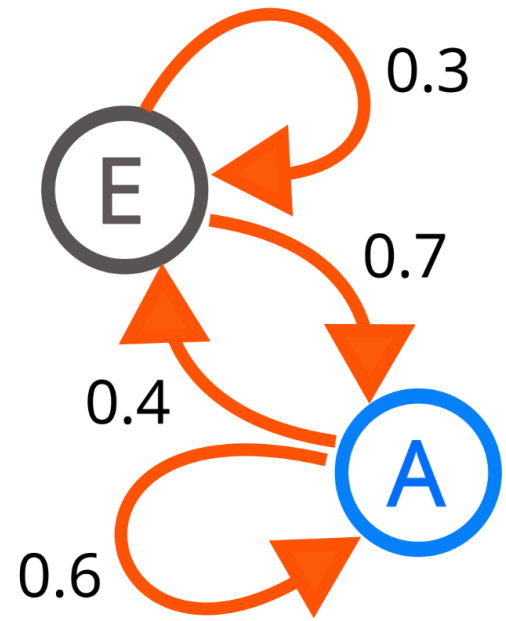
References

- S.M. Ross, “Stochastic Processes”: Chapter 4
- R. Wolff: Chapter 3
- S.M. Ross, “Introduction to Probability Models”: Chapter 4
- M. Harchol-Balter: Chapters 8 and 9

Discrete-Time Markov Chains

- Consider a stochastic process $\{X_n, n = 0,1,2, \dots\}$ that takes values from a countable state space S
- Unless otherwise mentioned, we assume that $S = \{0,1,2, \dots\}$
- If $X_n = i$, then process said to be in *state* i at time n
- $\{X_n\}$ said to be a discrete-time Markov chain (DTMC) if:
  - ❑ $P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0) = P(X_{n+1} = j | X_n = i)$ for all states $i_0, i_1, \dots, i_{n-1}, i, j$, and all $n \geq 0$
- E.g.:
  - ❑ $P(X_4 = 10 | X_3 = 5, X_2 = 12, X_1 = 3, X_0 = 7) = P(X_4 = 10 | X_3 = 5)$
- *Markovian Property*:
  - ❑ Given present state X_n, future state X_{n+1} is independent of past states $X_0, X_1, \dots, X_{n-1}$
- For $i, j \in \{0,1,2, \dots\}$, let $P_{ij} = P(X_{n+1} = j | X_n = i)$:
  - ❑ probability that when process is in state i, it will make a transition into state j
  - ❑ called *transition probability*



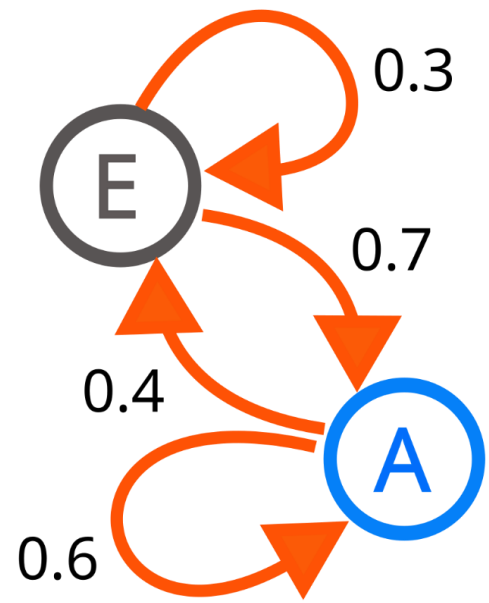
Ref: https://en.wikipedia.org/wiki/Markov_chain

Transition Probabilities

- Recall:
  - ❑ transition probabilities are $P_{ij} = P(X_{n+1} = j | X_n = i)$, where $i, j \in \{0,1,2,\dots\}$
- Properties:

  1) $P_{ij} \geq 0$, for all $i, j \in \{0,1,2,\dots\}$

  2) $\sum_{j=0}^{\infty} P_{ij} = 1$, for all $i \in \{0,1,2,\dots\}$

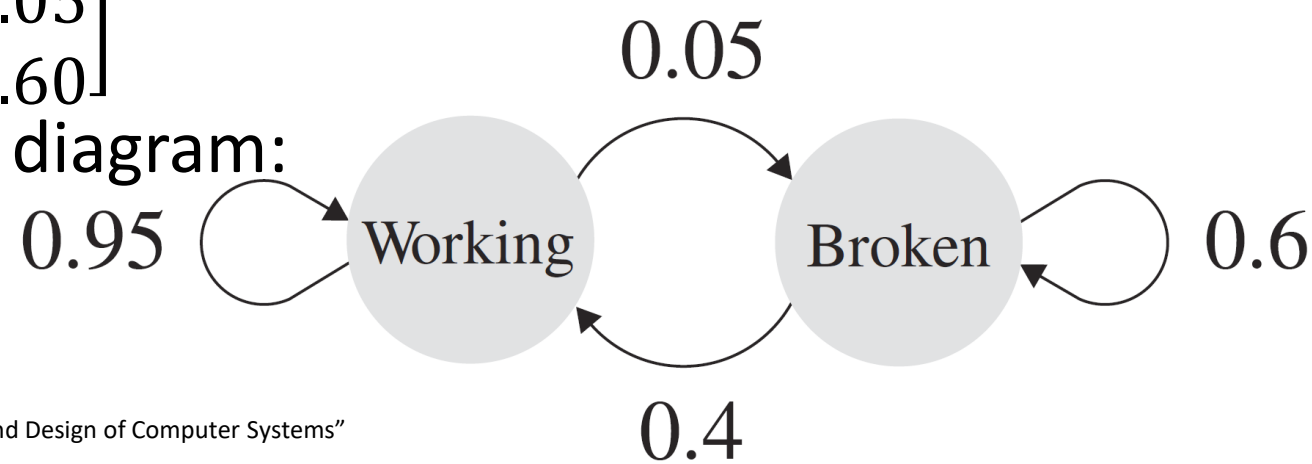


- Let P be matrix of transition probabilities:
  - ❑ $P = \begin{bmatrix} P_{00} & P_{01} & P_{02} & \cdots \\ P_{10} & P_{11} & P_{12} & \cdots \\ \vdots & & & \\ P_{i0} & P_{i1} & P_{i2} & \cdots \\ \vdots & & & \vdots \end{bmatrix}$
- If there are an infinite number of states in S, then P has infinite order
- P called *stochastic matrix*:
  - ❑ which is a matrix that satisfies 1) and 2)

Ref: https://en.wikipedia.org/wiki/Markov_chain

Example 1: Repair Facility Problem

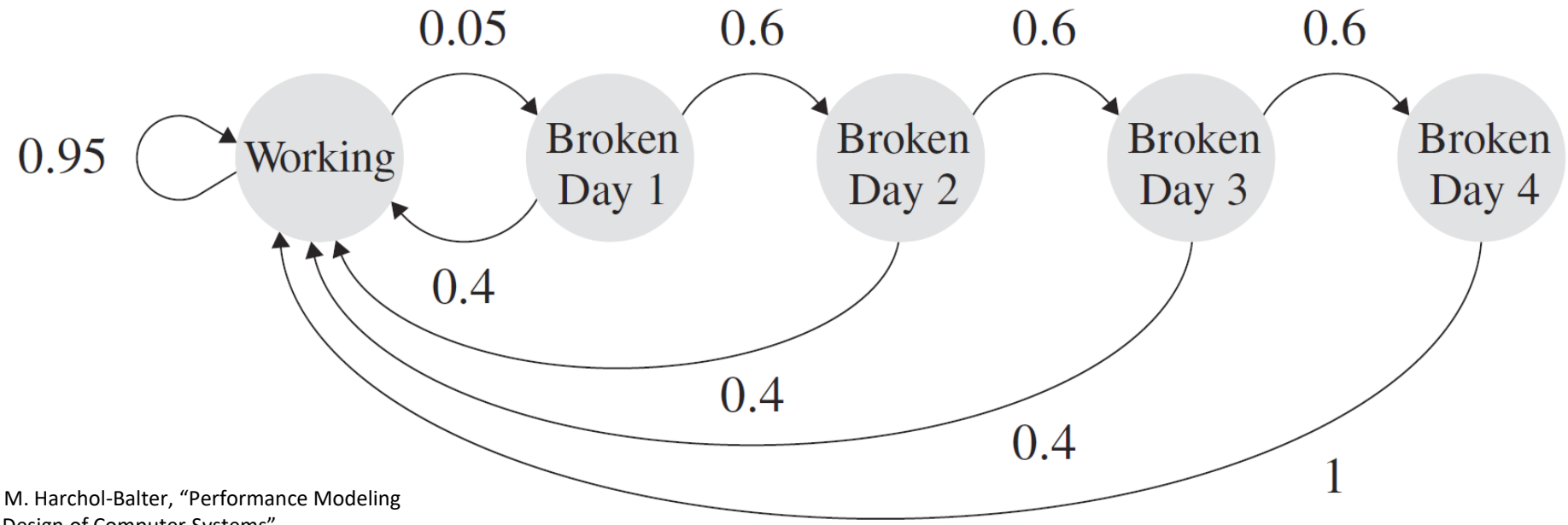
- A machine is either working or in repair center
- If it is working today, then there is a $95\ \%$ chance that it will be working tomorrow
- If it is in repair center today, then there is a $40\ \%$ chance that it will be working tomorrow
- States of DTMC:
  - 0: Working
  - 1: Broken (In repair center)
- Transition probability matrix:
  - $P = \begin{bmatrix} 0.95 & 0.05 \\ 0.40 & 0.60 \end{bmatrix}$
- State transition diagram:
  - shown in fig.



Ref: M. Harchol-Balter, "Performance Modeling and Design of Computer Systems"

Example 1: Repair Facility Problem (contd.)

- Now suppose that after machine remains broken for four days:
  - It is replaced with a new machine
- State transition diagram:
  - shown in fig.



Ref: M. Harchol-Balter, "Performance Modeling and Design of Computer Systems"

Example 2: The M/G/1 Queue

- Consider an M/G/1 queue with service distribution $G(\cdot)$
- Let $X(t)$ be number of customers in system at time t
- Does $\{X(t): t \geq 0\}$ have Markovian property?
  - ❑No, since future evolution of system depends on how long the customer in service has already been there
- Consider the stochastic process $X_1, X_2, \ldots$ obtained by looking at above system at instants when customers depart:
  - ❑X_n: no. of customers left behind by nth departure, for $n \geq 1$
- For $n \geq 1$, let Y_n: number of customers arriving during service period of $(n+1)$st customer
- X_{n+1} in terms of X_n:

1) $$\begin{cases} X_n - 1 + Y_n, & \text{if } X_n > 0, \\ Y_n, & \text{else.} \end{cases}$$

Example 2: The M/G/1 Queue (contd.)

- Recall that X_{n+1}:

  1) $\begin{cases} X_n - 1 + Y_n, & \text{if } X_n > 0, \\ Y_n, & \text{else.} \end{cases}$

- Distribution of Y_n, $n \geq 1$:

  2) $P\{Y_n = j\} = \int_0^\infty e^{-\lambda x} \frac{(\lambda x)^j}{j!} dG(x)$, for $j \in \{0,1,2,\dots\}$

  - also, Y_n, $n \geq 1$, are independent

- $\{X_n, n = 1,2,\dots\}$ is a DTMC with following transition probabilities:

  - $P_{0j} = \int_0^\infty e^{-\lambda x} \frac{(\lambda x)^j}{j!} dG(x)$, for $j \in \{0,1,2,\dots\}$
  - $P_{ij} = \begin{cases} \int_0^\infty e^{-\lambda x} \frac{(\lambda x)^{j-i+1}}{(j-i+1)!} dG(x), & \text{if } j \geq i-1, i \geq 1, \\ 0, & \text{else.} \end{cases}$

- Above DTMC is obtained by looking at original process only at certain time instants:

  - known as *embedded DTMC*

Example 3: The G/M/1 Queue

- Consider a G/M/1 queue such that customers arrive in accordance with a renewal process with interarrival distribution $G(\cdot)$
- Suppose service distribution is exponential with rate μ
- Let $X(t)$ be number of customers in system at time t
- Does $\{X(t): t \geq 0\}$ have Markovian property?
  - ❑No, since future evolution of system depends on amount of time that has elapsed since a customer arrived
- Consider the stochastic process $X_1, X_2, \ldots$ obtained by looking at above system at instants when customers arrive:
  - ❑X_n: number of customers in system as seen by nth arrival, for $n \geq 1$
- $\{X_n, n = 1,2, \ldots\}$ is a DTMC

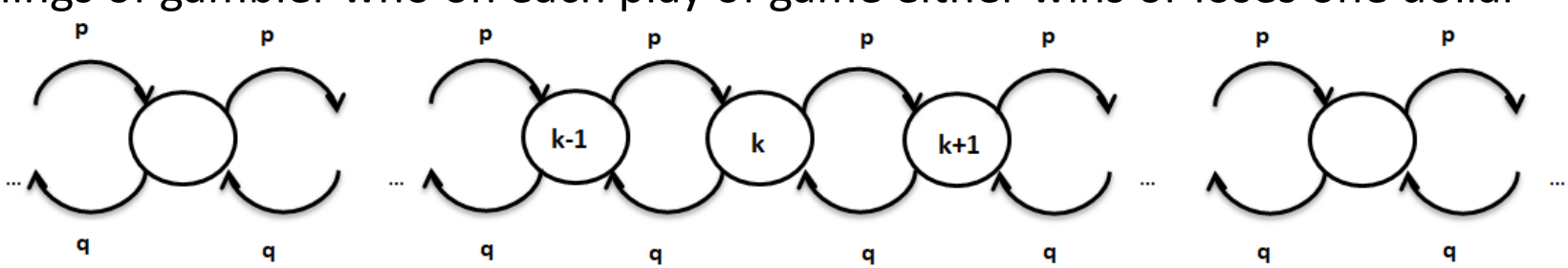
Example 3: The G/M/1 Queue (contd.)

- Distribution of number of customers served in an interval of length t, assuming that there are some customers to be served throughout the interval:
  - Poisson with mean μt
- $P_{i,i+1-j}$:
  - $\int_0^\infty e^{-\mu t} \frac{(\mu t)^j}{j!} dG(t)$, for $j \in \{0,1,\dots,i\}$
- $P_{i,0}$:
  - probability that at least $i+1$ Poisson events occur in random length of time having distribution G
  - $\int_0^\infty \left(\sum_{k=i+1}^\infty e^{-\mu t} \frac{(\mu t)^k}{k!} \right) dG(t)$, for $i \in \{0,1,\dots\}$
- Above DTMC is another example of an embedded DTMC

Example 4: Random Walk

- Let X_i, $i \geq 1$, be i.i.d. with:
  - $P\{X_i = j\} = a_j$, for $j \in \{0, \pm 1, \dots\}$
- Let:
  - $S_0 = 0$,
  - $S_n = \sum_{i=1}^{n} X_i$
- Then $\{S_n, n = 0,1,2, \dots\}$:
  - is a DTMC with transition probabilities $P_{ij} = a_{j-i}$
  - called a *general random walk*
- Following special case called *simple random walk*:
  - $a_j = \begin{cases} p, & \text{if } j = 1, \\ q, & \text{if } j = -1, \\ 0, & \text{else.} \end{cases}$
  - where $0 < p < 1$ and $q = 1 - p$
- Interpretation:
  - wanderings of drunken person as he/ she walks along straight line, or
  - winnings of gambler who on each play of game either wins or loses one dollar

Ref: https://www.analyticsvidhya.com/blog/2021/02/markov-chain-mathematical-formulation-intuitive-explanation-applications/



Chapman-Kolmogorov Equations

- For $i, j \geq 0$ and $n \geq 0$, let:
  - ❑ $P_{ij}^{n} = P\{X_{n+m} = j | X_m = i\}$,
  - ❑ called n-step transition probabilities
- Note that $P_{ij}^{1} = P_{ij}$ for all $i, j \geq 0$
- Suppose we know P_{ij}^{n} and P_{ij}^{m} for some $m, n \geq 0$ and all $i, j \geq 0$
- Then P_{ij}^{n+m}:
  - ❑ $P\{X_{n+m} = j | X_0 = i\}$
  - ❑ $\sum_{k=0}^{\infty} P\{X_{n+m} = j | X_n = k, X_0 = i\} P\{X_n = k | X_0 = i\}$
  - ❑ $\sum_{k=0}^{\infty} P_{kj}^{m} P_{ik}^{n}$
- Hence:

  1) $P_{ij}^{n+m} = \sum_{k=0}^{\infty} P_{ik}^{n} P_{kj}^{m}$, for all $m, n \geq 0$ and all $i, j \geq 0$
- Equations 1) known as *Chapman-Kolmogorov equations*
- Let $P^{(n)}$ be matrix of n-step transition probabilities P_{ij}^{n}
- Then $P^{(n+m)}$:
  - ❑ $P^{(n)} P^{(m)}$
- Hence, for each $n \geq 1$, $P^{(n)}$:
  - ❑ P^{n}

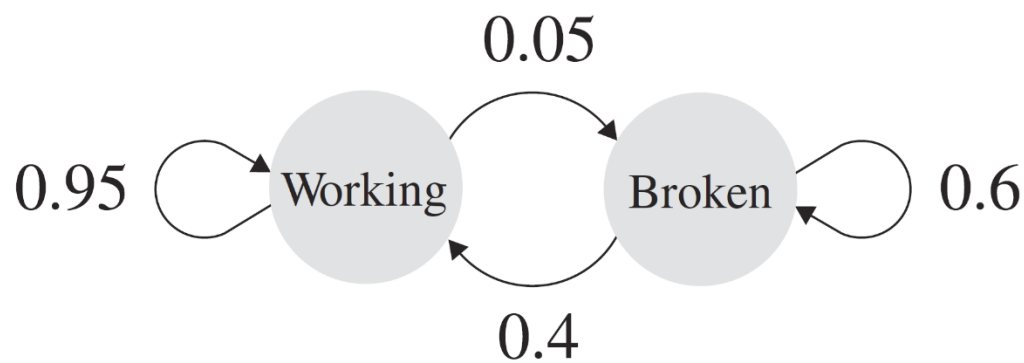
Example: Repair Facility Problem (contd.)

- States of DTMC:
  - ❑ 0: Working
  - ❑ 1: Broken (In repair center)
- General transition probability matrix:
  - ❑ $P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}$,
  - ❑ where $0 < a < 1$ and $0 < b < 1$
- P^n:

$$1) \quad \begin{bmatrix} \frac{b+a(1-a-b)^n}{a+b} & \frac{a-a(1-a-b)^n}{a+b} \\ \frac{b-b(1-a-b)^n}{a+b} & \frac{a+b(1-a-b)^n}{a+b} \end{bmatrix}$$

  - ❑ **Exercise**: Prove 1) by induction
- $\lim_{n\to\infty} P^n$:

$$2) \quad \begin{bmatrix} \frac{b}{a+b} & \frac{a}{a+b} \\ \frac{b}{a+b} & \frac{a}{a+b} \end{bmatrix}$$



- Note that all rows in 2) are the same
- Later we find general conditions under which this is true for a DTMC

Ref: M. Harchol-Balter, "Performance Modeling and Design of Computer Systems"

Communication

- State j is said to be accessible from state i if for some $n \geq 1$, $P_{ij}^{n} > 0$:
  - alternatively, we say state i can reach state j
  - denoted by $i \to j$
- $i \to j$ iff:
  - there is a path (finite sequence of transitions) of positive probability that starts in i and ends in j
- Two states i and j accessible to each other are said to *communicate* and we write:
  - $i \leftrightarrow j$
- By definition, we say that for every state i:
  - $i \leftrightarrow i$
  - i.e., every state communicates with itself
- **Proposition**: Communication is an equivalence relation, i.e., for states i, j, k:

  i. $i \leftrightarrow i$;

  ii. If $i \leftrightarrow j$, then $j \leftrightarrow i$;

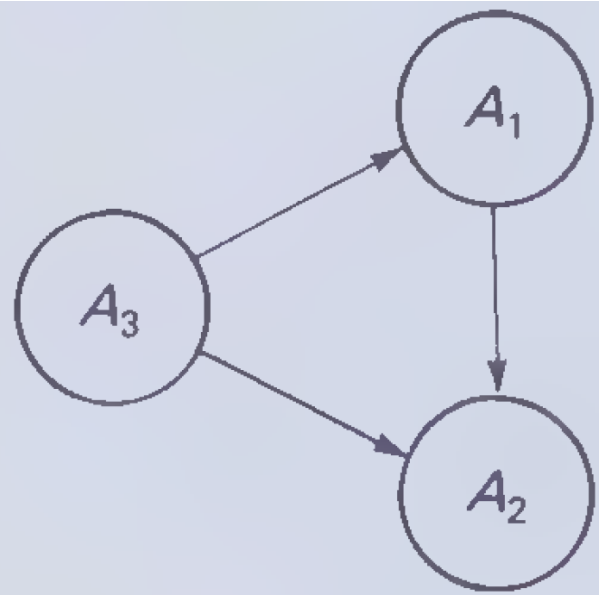
  iii. If $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$.
- **Proof**:
  - First two parts trivially follow from definition
  - To prove iii., suppose there exist m and n such that $P_{ij}^{m} > 0$ and $P_{jk}^{n} > 0$
  - Then P_{ik}^{m+n}:
    - $\sum_{r=0}^{\infty} P_{ir}^{m} P_{rk}^{n} \geq P_{ij}^{m} P_{jk}^{n} > 0$

Communication (contd.)

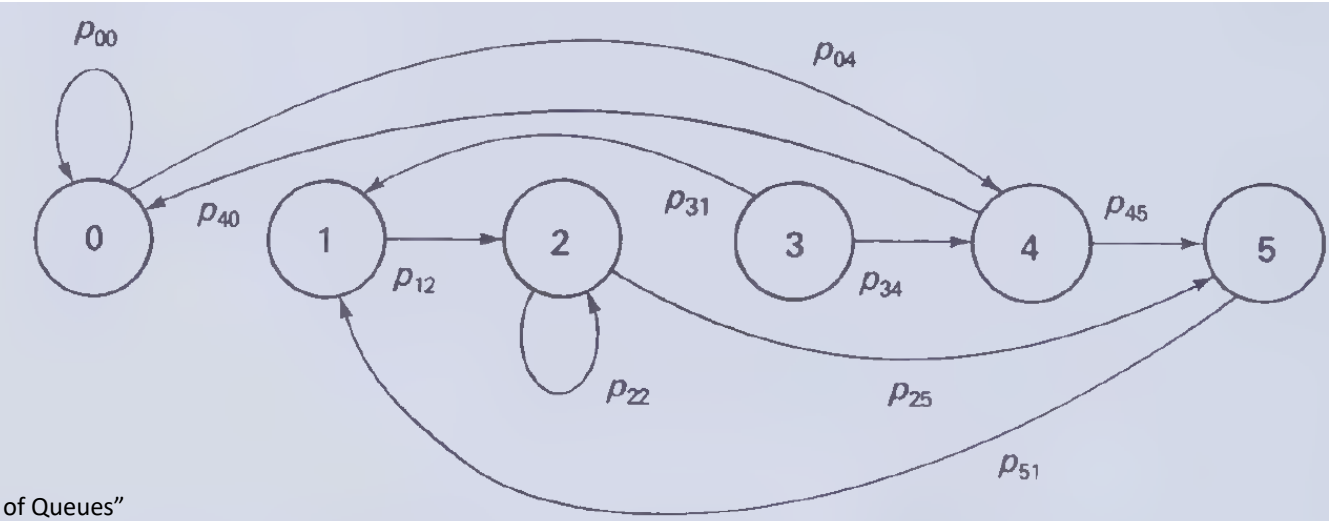
- Recall:
  - **Proposition**: Communication is an equivalence relation, i.e., for states i, j, k:
    - *i.* $i \leftrightarrow i$;
    - ii. If $i \leftrightarrow j$, then $j \leftrightarrow i$;
    - iii. If $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$.
- Two states that communicate are said to be in same *communication class*
- Any two classes are either disjoint or identical:
  - this follows from above Proposition
- We can partition the set of states S of the DTMC into communication classes $S_1, S_2, \ldots$ with the following properties:
  - a) Every state i belongs to exactly one class, say S_k,
  - b) Every class S_k contains at least one state,
  - c) States in the same class communicate with each other,
  - d) States belonging to different classes do not communicate
- The DTMC is said to be *irreducible* if:
  - there is only one class, i.e., all states communicate with each other

Example

- Fig. below shows:
  - a transition probability matrix, where $(*)$ denotes the position of each positive element, and
  - state transition diagram corresponding to this matrix
- What are the communication classes of this DTMC?
- $0 \leftrightarrow 4$:
  - since $0 \to 4$ and $4 \to 0$
- Also, $5 \leftrightarrow 1 \leftrightarrow 2$:
  - since $5 \to 1 \to 2 \to 5$
- No state can reach state 3
- No state in $\{5,1,2\}$ can reach $\{0,4\}$
- Hence, the communication classes are:
  - $A_1 = \{0,4\}$
  - $A_2 = \{5,1,2\}$
  - $A_3 = \{3\}$
- "Can reach" relations between different classes:
  - are shown in fig. on right



$$P = \begin{pmatrix} * & 0 & 0 & 0 & * & 0 \\ 0 & 0 & * & 0 & 0 & 0 \\ 0 & 0 & * & 0 & 0 & * \\ 0 & * & 0 & 0 & * & 0 \\ * & 0 & 0 & 0 & 0 & * \\ 0 & * & 0 & 0 & 0 & 0 \end{pmatrix}$$



Ref: R. Wolff, "Stochastic Modeling and the Theory of Queues"

Recurrence and Transience

- For any states i and j, let:
  - $f_{ij}^{n} = P\{X_n = j, X_k \neq j, k = 1, \dots, n-1 | X_0 = i\}$ for $n \geq 1$
  - probability that, starting in i, first transition into j occurs at time n
- As a convention, we let $f_{ij}^{0} = 0$ for all i and j
- Probability of ever making a transition into state j, given that process starts in i:
  - $f_{ij} = \sum_{n=1}^{\infty} f_{ij}^{n}$
- For $i \neq j$, $f_{ij} > 0$ iff:
  - j is accessible from i
- State j is said to be:
  - *recurrent* if $f_{jj} = 1$
  - *transient* if $f_{jj} < 1$

A Necessary and Sufficient Condition for Recurrence

- **Proposition**:
  - ❑ State j is recurrent iff:
    - $\sum_{n=1}^{\infty} P_{jj}^{n} = \infty$
- **Proof**:
  - ❑ State j is recurrent if, w.p. 1, process starting at j will eventually return
  - ❑ By Markovian property, process probabilistically restarts itself upon returning to j
  - ❑ Hence, if j is recurrent and $X_0 = j$, then the number of visits to j:
    - will be infinite w.p. 1 and hence will have mean ∞
  - ❑ If j is transient and $X_0 = j$, then the number of visits to j:
    - will be geometric with finite mean $\frac{f_{jj}}{1-f_{jj}}$
  - ❑ Thus, state j is recurrent iff:
    - $E(\text{number of visits to } j | X_0 = j) = \infty$
  - ❑ Now, $E(\text{number of visits to } j | X_0 = j)$:
    - $E\left(\sum_{n=1}^{\infty} 1_{\{X_n = j\}} \middle| X_0 = j\right)$
    - $\sum_{n=1}^{\infty} E\left(1_{\{X_n = j\}} \middle| X_0 = j\right)$
    - $\sum_{n=1}^{\infty} P_{jj}^{n}$
- Above proof also shows that w.p. 1, a transient state will only be visited a finite number of times:
  - ❑ hence the name “transient”

Recurrence and Transience Are Class Properties

- **Lemma**:
  - If i is recurrent and $i \leftrightarrow j$, then j is recurrent
- **Proof**:
  - Let m and n be such that $P_{ij}^{n} > 0$ and $P_{ji}^{m} > 0$
  - For any $s \geq 0$:
    - $P_{jj}^{m+n+s} \geq P_{ji}^{m} P_{ii}^{s} P_{ij}^{n}$
  - So, $\sum_{s=1}^{\infty} P_{jj}^{m+n+s} \geq$:
    - $P_{ji}^{m} P_{ij}^{n} \sum_{s=1}^{\infty} P_{ii}^{s} = \infty$
  - Hence, result follows from above Proposition
- Above lemma implies following result
- **Theorem**: The states in a communication class A are either:
  - all recurrent, or
  - all transient

Example

- Recall simple random walk:
  - ❑ State space is set of all integers $\{0, \pm 1, \dots\}$
  - ❑ Transition probabilities $P_{i,i+1} = p$ and $P_{i,i-1} = 1 - p$ for all $i \in \{0, \pm 1, \dots\}$:
    - where $0 < p < 1$
- Want to find out which states are recurrent and which transient
- Since all states communicate, it follows from above Proposition that either all states are transient or all recurrent
- Consider state 0 and let us find:
  - ❑ whether $\sum_{n=1}^{\infty} P_{00}^{n}$ is finite or infinite
- Since starting from 0, process cannot return to 0 after odd number of steps:
  - ❑ $P_{00}^{2n+1} = 0$, for all $n \in \{1,2, \dots\}$
- Also, for each $n \in \{1,2, \dots\}$, P_{00}^{2n}:
  - ❑ $\binom{2n}{n} p^n (1-p)^n$
  - ❑ $\frac{(2n)!}{n!n!} \big(p(1-p)\big)^n$

Example (contd.)

- Recall: for each $n \in \{1,2,\dots\}$, P_{00}^{2n}:
  - $\frac{(2n)!}{n!n!}\left(p(1-p)\right)^n$
- By Stirling's approximation:
  - $n! \sim \left(n^{n+\frac{1}{2}}\right)e^{-n}\sqrt{2\pi}$
  - where $a_n \sim b_n$ if $\lim_{n\to\infty} \frac{a_n}{b_n} = 1$
- So, P_{00}^{2n}:
  - $\sim \frac{\left(4p(1-p)\right)^n}{\sqrt{\pi n}}$
- **Exercise**: Show that if $a_n \sim b_n$, then $\sum_{n=1}^{\infty} a_n < \infty$ iff $\sum_{n=1}^{\infty} b_n < \infty$
- Hence, $\sum_{n=1}^{\infty} P_{00}^{n} < \infty$ iff following quantity is finite:
  - $\sum_{n=1}^{\infty} \frac{\left(4p(1-p)\right)^n}{\sqrt{\pi n}}$
- Now, $4p(1-p) \leq 1$, with equality holding iff $p = \frac{1}{2}$
- So, $\sum_{n=1}^{\infty} P_{00}^{n} = \infty$ iff $p = \frac{1}{2}$
- Thus, the DTMC is:
  - recurrent if $p = \frac{1}{2}$
  - transient if $p \neq \frac{1}{2}$

Closed and Open Sets

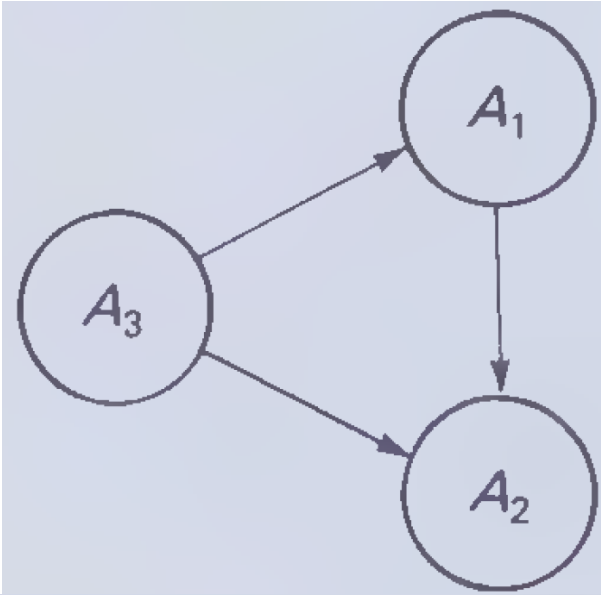
- A set of states C is said to be *closed* if:
  - For every pair of states $i \in C$ and $j \notin C$, $P_{ij} = 0$
- Clearly:
  - Paths of positive probability starting in C must remain in C
  - $P_{ij}^{(n)} = 0$ for every $n \geq 1$
- If C is not closed, it is said to be *open*

Transience of Open Communication Classes

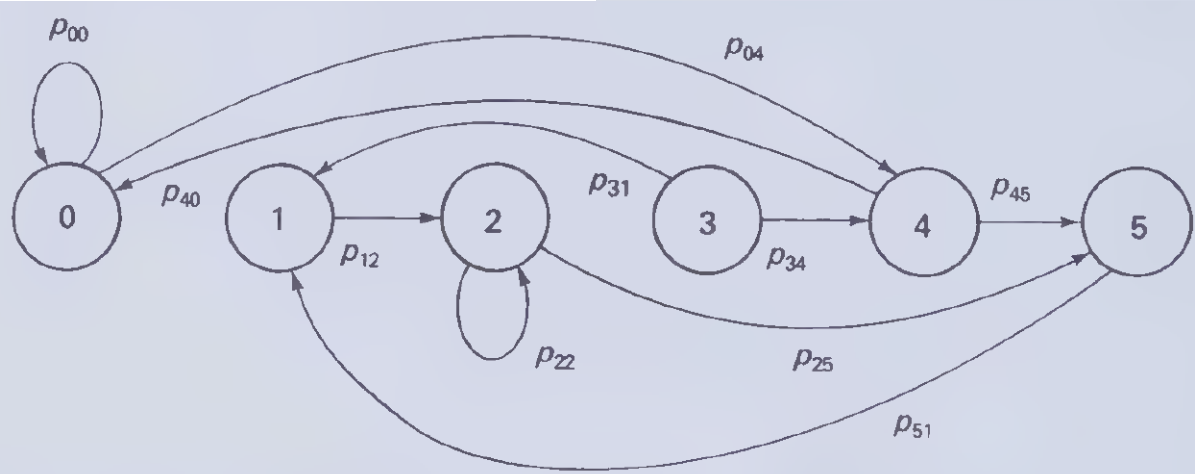
- **Theorem**: Open communication classes are transient
- **Proof**:
  - ❑Let A be an open communication class
  - ❑So, there exist states $i \in A$ and $j \notin A$ s.t. $P_{ij} > 0$
  - ❑Hence, $i \to j$, but $j \nrightarrow i$:
    - ○ otherwise j would be in A
  - ❑So, $f_{ji} = 0$
  - ❑Suppose $X_0 = i$
  - ❑Conditioning on whether the next transition is to state j:
    - ○ $f_{ii} = f_{ji}P_{ij} + P(\text{process returns to state } i | X_1 \neq j)\left(1 - P_{ij}\right)$
    - ○ $\leq 0 + \left(1 - P_{ij}\right) < 1$
  - ❑So, state i is transient

Recall: Example

- The communication classes are:
  - ❑ $A_1 = \{0,4\}$
  - ❑ $A_2 = \{5,1,2\}$
  - ❑ $A_3 = \{3\}$
- "Can reach" relations between different classes are shown in fig. on right
- Which classes are transient and which are recurrent?
- Classes A_1 and A_3 are open and hence transient
- Class A_2 is closed and finite; later, we will show that this implies it is recurrent



$$P = \begin{pmatrix} * & 0 & 0 & 0 & * & 0 \\ 0 & 0 & * & 0 & 0 & 0 \\ 0 & 0 & * & 0 & 0 & * \\ 0 & * & 0 & 0 & * & 0 \\ * & 0 & 0 & 0 & 0 & * \\ 0 & * & 0 & 0 & 0 & 0 \end{pmatrix}$$



Ref: R. Wolff, "Stochastic Modeling and the Theory of Queues"

Stationary Distribution

- Consider a DTMC with:
  - ❑ transition probabilities $P_{ij} = P(X_{n+1} = j | X_n = i)$, where $i, j \in \{0,1,2, \dots\}$, and
  - ❑ transition probability matrix P
- A probability distribution $\{\pi_j, j \geq 0\}$ is said to be *stationary* for the DTMC if:
  - ❑ $\pi_j = \sum_{i=0}^{\infty} \pi_i P_{ij}, \forall j \in \{0,1,2, \dots\}$
- Suppose the probability distribution of X_n is the stationary distribution, i.e.:
  - ❑ $P\{X_n = j\} = \pi_j, \forall j \in \{0,1,2, \dots\}$
- Then $P\{X_{n+1} = j\}$:
  - ❑ $\sum_{i=0}^{\infty} P\{X_{n+1} = j | X_n = i\} P\{X_n = i\}$
  - ❑ $\sum_{i=0}^{\infty} \pi_i P_{ij}$
  - ❑ π_j
- That is, X_{n+1} also has the stationary distribution
- Hence, if X_0 has the stationary distribution, then:
  - ❑ for all $n \geq 1$, X_n has the stationary distribution
- Also, by Markovian property, for each $m \geq 0$:
  - ❑ $X_n, X_{n+1}, \dots, X_{n+m}$ will have same joint distribution for each n
- That is:
  - ❑ $\{X_n, n \geq 0\}$ will be a stationary process

Stationary Distribution (contd.)

- Recall: a probability distribution $\{\pi_j, j \geq 0\}$ is said to be stationary for the DTMC if:

  1) $\pi_j = \sum_{i=0}^{\infty} \pi_i P_{ij}, \forall j \in \{0,1,2, \dots\}$

- Consider the row vector:
  - $\pi = [\pi_0, \pi_1, \dots]$
- Using 1), we can write:
  - $\pi = \pi P$

Example 1

- Suppose that:
  - if it rains today, then it will rain tomorrow w.p. α
  - if it does not rain today, then it will rain tomorrow w.p. β
- This is a DTMC with states:
  - 0 (it rains),
  - 1 (it does not rain)
- Transition probability matrix:
  - $P = \begin{bmatrix} \alpha & 1-\alpha \\ \beta & 1-\beta \end{bmatrix}$
- Stationary distribution obtained by solving following equations:
  - $\pi_0 = \alpha\pi_0 + \beta\pi_1,$
  - $\pi_1 = (1-\alpha)\pi_0 + (1-\beta)\pi_1,$
  - $\pi_0 + \pi_1 = 1$
- Solution:
  - $\pi_0 = \frac{\beta}{1+\beta-\alpha}, \pi_1 = \frac{1-\alpha}{1+\beta-\alpha}$
- E.g.: if $\alpha = 0.7$ and $\beta = 0.4$, then:
  - $\pi_0 = \frac{4}{7} \approx 0.571$

Example 2

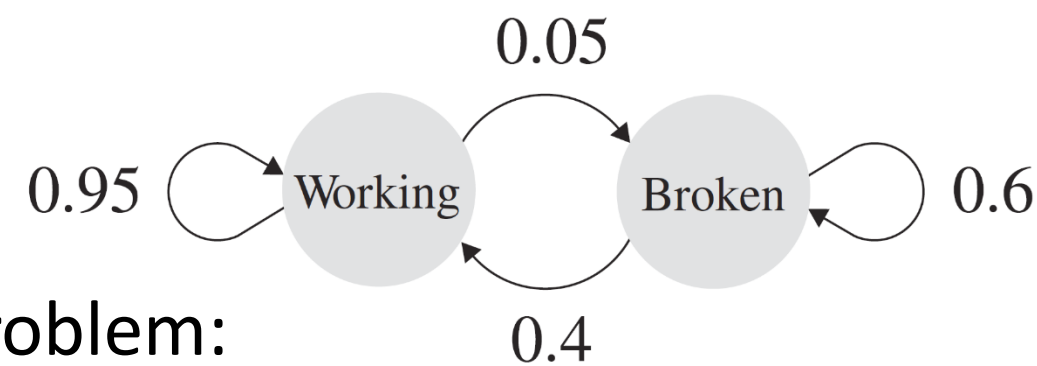
- Suppose transitions between social classes of successive generations in a family modeled as transitions of a DTMC:
  - i.e., we assume that occupation of a child depends only on his or her parent's occupation
- Assume that there are three social classes (states):
  - upper-class (0)
  - middle-class (1)
  - lower-class (2)
- Transition probability matrix is:
  - $P = \begin{bmatrix} 0.45 & 0.48 & 0.07 \\ 0.05 & 0.70 & 0.25 \\ 0.01 & 0.50 & 0.49 \end{bmatrix}$
- Stationary distribution obtained by solving following equations:
  - $\pi_0 = 0.45\pi_0 + 0.05\pi_1 + 0.01\pi_2,$
  - $\pi_1 = 0.48\pi_0 + 0.70\pi_1 + 0.50\pi_2,$
  - $\pi_2 = 0.07\pi_0 + 0.25\pi_1 + 0.49\pi_2,$
  - $\pi_0 + \pi_1 + \pi_2 = 1$
- Solution:
  - $\pi_0 = 0.07, \pi_1 = 0.62, \pi_2 = 0.31$

Periodicity

- For a state i, the period $d(i)$ is the greatest common divisor (GCD) of all possible numbers of steps n for which the probability of returning to state i is greater than zero
- That is:
  - $d(i) = \gcd\{n > 0 : P_{ii}^{n} > 0\}$
- If $P_{ii}^{n} = 0$ for all $n > 0$, then we define the period of i to be ∞
- A state i with period $d(i) = 1$ is said to be *aperiodic*

Examples

- Recall:
  - $d(i) = \gcd\{n > 0: P_{ii}^n > 0\}$
- E.g. 1: simple random walk:
  - State space is set of all integers $\{0, \pm 1, \dots\}$
  - Transition probabilities $P_{i,i+1} = p$ and $P_{i,i-1} = 1 - p$ for all $i \in \{0, \pm 1, \dots\}$:
    - where $0 < p < 1$
  - Period of state i:
    - $2, \forall i \in \{0, \pm 1, \dots\}$
- E.g. 2: repair facility problem:
  - State transition diagram shown in fig.
  - Period of state i:
    - 1, for both states
  - That is, the DTMC is aperiodic



Ref: M. Harchol-Balter, "Performance Modeling and Design of Computer Systems"

Periodicity (contd.)

- Recall:
  - $d(i) = \gcd\{n > 0 : P_{ii}^n > 0\}$
- Following result shows that periodicity is a class property
- **Proposition**: If $i \leftrightarrow j$, then $d(i) = d(j)$
- **Proof**:
  - Let m, n, and s be such that $P_{ij}^m P_{ji}^n > 0$ and $P_{ii}^s > 0$
  - Then P_{jj}^{n+m}:
    - $\geq P_{ji}^n P_{ij}^m > 0$
  - Also, P_{jj}^{n+s+m}:
    - $\geq P_{ji}^n P_{ii}^s P_{ij}^m > 0$
  - So, $d(j)$ divides both $n + m$ and $n + s + m$ and hence divides s
  - So, $d(j)$ divides $d(i)$
  - By symmetry, $d(i)$ divides $d(j)$

First Passage Time

- Recall: for any states i and j:
  - $f_{ij}^{n} = P\{X_n = j, X_k \neq j, k = 1, \dots, n-1 | X_0 = i\}$ for $n \geq 1$
  - probability that, starting in i, first transition into j occurs at time n
- Let T_{ij}: time at which first transition into state j occurs, given that the process starts in i:
  - $T_{ij} = \min\{n \geq 1 : X_n = j\}$ given that $X_0 = i$
  - known as *first passage time*
- $P\left(T_{ij} = n\right)$:
  - f_{ij}^{n}
- For $i = j$, a visit is called a *return*:
  - T_{jj}: first return time to state j
- State j is called *absorbing* if $P_{jj} = 1$

Example

- Consider a DTMC with transition probability matrix:
  - $P = \begin{bmatrix} a & b & c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$,
  - where $abc > 0$
- Which states are absorbing?
  - states 1 and 2
- For $n \geq 1$, f_{01}^n:
  - $a^{n-1}b$
- For $n \geq 1$, f_{02}^n:
  - $a^{n-1}c$
- f_{01}:
  - $\sum_{n=1}^{\infty} f_{01}^n$
  - $\frac{b}{1-a}$
- f_{02}:
  - $\sum_{n=1}^{\infty} f_{02}^n$
  - $\frac{c}{1-a}$
- f_{00}:
  - $\sum_{n=1}^{\infty} f_{00}^n$
  - a, since $f_{00}^1 = a$ and $f_{00}^n = 0$ for $n \geq 2$
- So, $f_{01} + f_{02} = 1$:
  - consistent with fact that w.p. 1, process either reaches state 1 or 2, but not both

Probability of Ever Reaching a Recurrent State

- Recall:
  - $f_{ij}^n = P\{X_n = j, X_k \neq j, k = 1, \dots, n-1 | X_0 = i\}$ for $n \geq 1$
  - $f_{ij} = \sum_{n=1}^{\infty} f_{ij}^n$
- **Proposition**: If $i \leftrightarrow j$ and j is recurrent, then $f_{ij} = 1$
- **Proof**:
  - Suppose $X_0 = i$, and let n be s.t. $P_{ij}^n > 0$
  - Say that we miss opportunity 1 if $X_n \neq j$:
    - If this happens, let T_1 denote the next time we enter i (note that T_1 is finite w.p. 1 since i is recurrent)
  - Say that we miss opportunity 2 if $X_{T_1+n} \neq j$:
    - If this happens, let T_2 denote the next time we enter i
  - Say that we miss opportunity 3 if $X_{T_2+n} \neq j$, and so on
  - Opportunity number of first success:
    - geometric r.v. with mean $\frac{1}{P_{ij}^n}$,
    - is hence finite w.p. 1

Positive Recurrent and Null Recurrent States

- Recall:
  - T_{ij}: first passage time
  - State j is said to be recurrent if $f_{jj} = 1$ and transient if $f_{jj} < 1$
- Let $\mu_{ij} = E(T_{ij})$
- μ_{jj}:
  - $\begin{cases} \infty, & \text{if } j \text{ is transient,} \\ \sum_{n=1}^{\infty} n f_{jj}^{n}, & \text{if } j \text{ is recurrent} \end{cases}$
- If state j is recurrent, then we say that it is:
  - *positive recurrent* if $\mu_{jj} < \infty$
  - *null recurrent* if $\mu_{jj} = \infty$

DTMC as a Renewal Process

- Let $N_j(t)$: number of visits to state j until time t
- Suppose j is recurrent and $X_0 = j$
- Then since process probabilistically starts over upon each transition into j:
  - $\{N_j(t), t \geq 1\}$ is an ordinary renewal process with interarrival distribution $\{f_{jj}^n, n \geq 1\}$
- If $X_0 = i$, $i \leftrightarrow j$, and j is recurrent, then:
  - $\{N_j(t), t \geq 1\}$ is a delayed renewal process with initial interarrival distribution $\{f_{ij}^n, n \geq 1\}$

Rate of Visits to State j

- Recall:
  1) State j is recurrent iff $\sum_{n=1}^{\infty} P_{jj}^{n} = \infty$
- Suppose a DTMC starts in state $X_0 = i$, and we are interested in counting visits to state j
- Similar to proof of 1), it can be shown that if state j is transient, then:
  - ❑ $\sum_{n=1}^{\infty} P_{ij}^{n} < \infty$, for all i
  - ❑ i.e., starting in state i, the expected number of visits to state j is finite
- So, for transient states j:
  - ❑ $\lim_{n\to\infty} P_{ij}^{n} = 0$
- Let $M_j(t) = E\{N_j(t)\}$:
  - ❑ $\sum_{n=1}^{t} P_{ij}^{n}$
  - ❑ expected number of visits to state j until time t

Rate of Visits to State j (contd.)

- Recall:
  - A DTMC starts in state $X_0 = i$, and we are interested in counting visits to state j
  - $M_j(t) = E\{N_j(t)\} = \sum_{n=1}^{t} P_{ij}^n$
    - expected number of visits to state j until time t
- **Theorem**: If $i \leftrightarrow j$, then:

$$a) \quad P\left\{\lim_{t\to\infty} \frac{N_j(t)}{t} = \frac{1}{\mu_{jj}} \middle| X_0 = i\right\} = 1$$

$$b) \quad \lim_{t\to\infty} \frac{M_j(t)}{t} = \lim_{t\to\infty} \frac{\sum_{n=1}^{t} P_{ij}^n}{t} = \frac{1}{\mu_{jj}}$$

- **Proof**:
  - parts a) and b) follow from strong law for renewal processes and elementary renewal theorem, respectively
- Note that:
  - $\lim_{t\to\infty} \frac{\sum_{n=1}^{t} P_{ij}^n}{t} = \frac{1}{\mu_{jj}} \begin{cases} = 0, & \text{if } j \text{ is null recurrent,} \\ > 0, & \text{if } j \text{ is positive recurrent} \end{cases}$

Positive Recurrence, Null Recurrence, and Transience Are Class Properties

- Recall:

  1) $\lim_{t\to\infty} \frac{\sum_{n=1}^{t} P_{ij}^{n}}{t} = \frac{1}{\mu_{jj}} \begin{cases} = 0, & \text{if } j \text{ is null recurrent,} \\ > 0, & \text{if } j \text{ is positive recurrent} \end{cases}$

  - ❑ **Lemma**:
    - If i is recurrent and $i \leftrightarrow j$, then j is recurrent
    - **Proof**:
      - Let m and n be such that $P_{ij}^{n} > 0$ and $P_{ji}^{m} > 0$
      - For any $s \geq 0$:

        2) $P_{jj}^{m+n+s} \geq P_{ji}^{m} P_{ii}^{s} P_{ij}^{n}$
      - So, $\sum_{s=1}^{\infty} P_{jj}^{m+n+s} \geq$:
        - $P_{ji}^{m} P_{ij}^{n} \sum_{s=1}^{\infty} P_{ii}^{s} = \infty$
      - Hence, j is recurrent

- **Lemma**:
  - ❑ If $i \leftrightarrow j$, then i is positive (respectively, null) recurrent iff j is positive (respectively, null) recurrent

- **Proof**:
  - ❑From 2), we get that:

    3) $\frac{\sum_{s=1}^{t} P_{jj}^{m+n+s}}{t} \geq \frac{P_{ji}^{m} P_{ij}^{n} \sum_{s=1}^{t} P_{ii}^{s}}{t}$
  - ❑ If j is null recurrent, then by 1), it follows that $\lim_{t\to\infty} \frac{\sum_{s=1}^{t} P_{jj}^{m+n+s}}{t} = 0$
  - ❑So, by 3), $\lim_{t\to\infty} \frac{\sum_{s=1}^{t} P_{ii}^{s}}{t} = 0$ and hence i is null recurrent

- **Theorem**: The states in a communication class A are either:
  - ❑all positive recurrent,
  - ❑all null recurrent, or
  - ❑all transient

Positive Recurrence of Finite Closed Communication Classes

- **Theorem**: Finite closed communication classes are positive recurrent
- **Proof**:
  - Let A be a finite closed communication class
  - For all states $i \in A$:
    1) $\sum_{j \in A} P_{ij}^{n} = 1$, for all $n = 1,2, \ldots$
  - If A is either transient or null recurrent, then for all $j \in A$:
    2) $\lim_{t \to \infty} \frac{\sum_{n=1}^{t} P_{ij}^{n}}{t} = 0$
  - Summing 2) over all $j \in A$:
    - $\sum_{j \in A} \left\{ \lim_{t \to \infty} \frac{\sum_{n=1}^{t} P_{ij}^{n}}{t} \right\} = 0$
  - However, by 1):
    - $\sum_{j \in A} \left\{ \lim_{t \to \infty} \frac{\sum_{n=1}^{t} P_{ij}^{n}}{t} \right\} = \lim_{t \to \infty} \sum_{n=1}^{t} \left(\frac{\sum_{j \in A} P_{ij}^{n}}{t} \right) = 1$
  - This is a contradiction

Example

- Consider the DTMC with states $0,1,2,3,4$ and following transition probability matrix:
  - $P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & 0 & 0 & \frac{1}{2} \end{bmatrix}$
- Communication classes:
  - $\{0,1\}$
  - $\{2,3\}$
  - $\{4\}$
- Which of the above classes are transient, positive recurrent, and null recurrent?
- $\{0,1\}$ and $\{2,3\}$ are positive recurrent
- $\{4\}$ is transient

Some Properties of Finite DTMCs

- **Theorem**:
  - For a finite DTMC, state i is transient iff there exists a state j s.t. $i \to j$, but $j \nrightarrow i$
  - Otherwise, state i is positive recurrent
- **Proof**:
  - Suppose there exists a state j s.t. $i \to j$, but $j \nrightarrow i$
  - Then class of state i is open and hence it is transient
  - To prove converse, suppose state i is transient
  - Assume that for every state j s.t. $i \to j$, it is the case that $j \to i$
  - Then class of state i is closed; since it is also finite, it is positive recurrent, which is a contradiction
- **Corollary**:
  - Null recurrent states cannot occur in finite DTMCs
  - Hence, null communication classes are infinite

Some Properties of Finite DTMCs (contd.)

- **Proposition**:
  - A finite DTMC must contain at least one closed (and hence positive recurrent) communication class
- **Proof**:
  - Start with any class, say A_1
  - If it is open, consider a class, say A_2 that can be reached from A_1 s.t. there is no path from A_2 to A_1
  - If A_2 is open, consider a class A_3 that can be reached from A_2 s.t. there is no path from A_3 to A_2
  - Continue this process
  - We can never return to a class considered earlier
  - Since the number of classes is finite, we will eventually reach a class that cannot reach any other class, i.e., it is closed

Probability Vectors and Transition Probability Matrices

- For $j \geq 0$:
  - Let $P(X_0 = j) = a_j$
  - For $n \geq 1$, let $P(X_n = j) = a_j^{(n)}$
- Consider the following row vectors:
  - $a = [a_0, a_1, \dots]$
  - $a^{(n)} = \left[a_0^{(n)}, a_1^{(n)}, \dots\right]$
- a and $a^{(n)}$ are called *probability vectors* since their elements are non-negative and sum to 1
- Suppose transition probability matrix is P
- $a^{(n)}$:
  - $a^{(n-1)}P$
  - aP^n